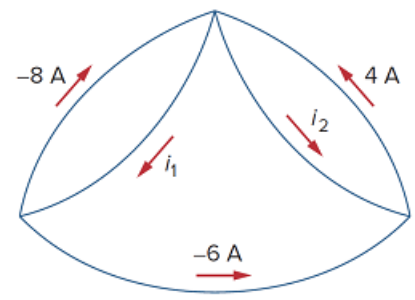


Problem

Determine i_1 and i_2 in the circuit of Fig. 2.74.

Figure 2.74:



Step-by-step solution

Step 1 of 3

The circuit is shown in Figure 1.

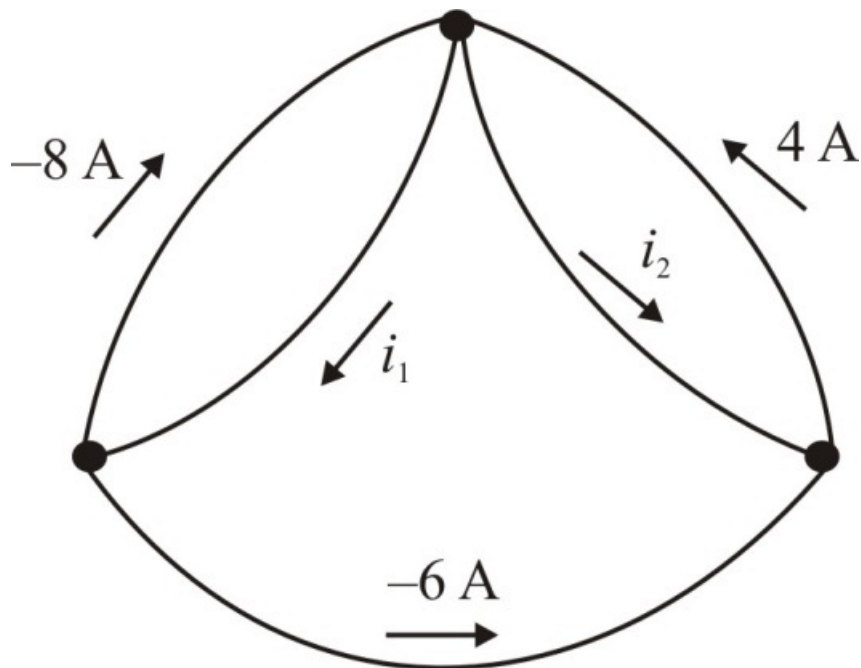


Figure 1

Step 2 of 3

Apply Kirchhoff's current law; the current entering any node is equal to the current leaving from that node. Calculate the current i_1 .

$$\begin{aligned} i_1 &= (-8) + (-6) \\ &= -14 \text{ A} \end{aligned}$$

Therefore, the current i_1 is -14 A.

Step 3 of 3

Apply Kirchhoff's current law; the current entering any node is equal to the current leaving from that node. Calculate the current i_2 .

$$i_2 + (-6) = 4$$

$$i_2 = 10 \text{ A}$$

Therefore, the current i_2 is 10 A.